

Contents

1	Basic	
1.1	default	
1.2	奇妙的東西	
1.3	random	
2	Binary Search	
2.1	二分搜	
2.2	三分搜	
3	DataStructure	
3.1	DSU(並查集)	
3.2	FenwickTree/BIT	
3.3	二維 BIT	
3.4	Seg/線段樹	
4	Graph	
4.1	Topo/拓撲	
4.2	bellman/邊長鬆弛	
4.3	ola/歐拉	
4.4	LCA/最近公共祖先	
4.5	km/二分圖最大權匹配	
4.6	Scs/強連通分量	
4.7	Bcc/雙連通分量	
5	dynamic programming	
5.1	SOS DP	
6	Computational Geometry	
6.1	default	
6.2	極座標排序	
6.3	ConvexHull/凸包	
6.4	PtSegDist	
6.5	點是否在多邊形內	
6.6	最遠點對/選轉卡尺	
7	Flow	
7.1	Dinic	
8	String	
8.1	KMP	
8.2	Z-value	
8.3	馬拉車	
8.4	minRotation/最小字典序旋轉	
8.5	SAIS	
8.6	Trie/字典樹	
8.7	ACAutomata/AC 自動機	
8.8	差分約束系統	
9	酷東西可能會用到	
9.1	areaofrect/矩形聯集面積	
10	Math	
10.1	basic	
10.2	快速冪	
10.3	大數運算	
10.4	Prime	
10.5	ExGCD	
10.6	Phi	
10.7	Euler/歐拉函數	
10.8	millerrabin/判斷質數	
10.9	Pollard Rho/質因數分解	
10.10	FFT/傅立葉轉換 (多項式相乘)	
10.11	Theorem	

1 Basic

1.1 default

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
#define idonthevegirlfriend ios::sync_with_stdio(0),cin.tie(0);
#define pb push_back
#define endl "\n"
#define all(v) (v).begin(),(v).end()
void solve(){
}
signed main(){
    idonthevegirlfriend
    int _=1;
    //cin>>_;
    while(--){
        solve();
    }
}
```

1.2 奇妙的東西

```
1 priority_queue<T> //由大到小
1 priority_queue<T,vector<T>,greater<T>> //由小到大
1 __builtin_popcountll // 二進位有幾個1(記得這是long long)
1 __builtin_clzll // 左起第一個1之前0的個數
1 __builtin_parityll // 1的個數的奇偶性
1 __builtin_mul_overflow(a,b,&h) // a*b是否溢位,h = a * b
2 ;
2 __builtin_add_overflow(a,b,&h)
```

1.3 random

```
mt19937 mt(chrono::steady_clock::now().time_since_epoch
().count());
//mt19937_64 mt() -> return random
int randint(int l, int r){
    uniform_int_distribution<> dis(l, r); return dis(mt
);
}
```

2 Binary Search

2.1 二分搜

```
int bsearch_1(int l, int r)
{
    while (l < r)
    {
        int mid = l + r >> 1;
        if (check(mid)) r = mid;
        else l = mid + 1;
    }
    return l;
}
// .....0000000000
int bsearch_2(int l, int r)
{
    while (l < r)
    {
        int mid = l + r + 1 >> 1;
        if (check(mid)) l = mid;
        else r = mid - 1;
    }
    return l;
}
// 000000000.....
int m = *ranges::partition_point(views::iota(0LL,(int)1
e9+9),[&](int a){
    return check(a) > k;
});
//[begin,last)
//1111111000000000000
//搜左邊數過來第一個 0
//都是 1 會回傳 last
int partitionpoint(int L,int R,function<bool(int)> chk)
{
    int l = L,r = R-1;
    while(r - l > 10){
        int m = l + (r-l)/2;
        if(chk(m)) l = m;
        else r = m;
    }
    int m = l;
    while(m <= r){
        if(!chk(m)) break;
        ++m;
    }
    if(!chk(m)) return m;
    else return R;
}
//手工
```

2.2 三分搜

```
int l = 1,r = 100;
while(l < r) {
    int lmid = l + (r - l) / 3; // l + 1/3區間大小
    int rmid = r - (r - l) / 3; // r - 1/3區間大小
    lans = cal(lmid),rans = cal(rmid);
```

```
// 求凹函數極小直
if(lans <= rans) r = rmid - 1;
else l = lmid + 1;
}
```

3 DataStructure

3.1 DSU(並查集)

```
struct DSU {
    vector<int> f, sz;
    int cnt;
    DSU(int n) : f(n), sz(n) {
        for (int i = 0; i < n; i++) {
            f[i] = i;
            sz[i] = 1;
        }
    }
    int find(int x) {
        if (x == f[x]) return x;
        return f[x] = find(f[x]);
    }
    void merge(int x, int y) {
        x = find(x); y = find(y);
        if (x == y) return;
        if (sz[x] < sz[y])
            swap(x, y);
        f[y] = x;
        sz[x] += sz[y];
        cnt--;
    }
    bool same(int a, int b) {
        return find(a) == find(b);
    }
    int groups() { // 找有幾群
        return cnt;
    }
};
```

3.2 FenwickTree/BIT

```
struct Fenwick { // 這個是1base的
    int n;
    vector<int> s;
    int lowbit(int x) { return x & -x; }
    Fenwick(int _n) {
        n = _n + 1;
        s.clear(), s.resize(n, 0);
    }
    void update(int i, int v) { // 更新數值
        for (; i < n; i += lowbit(i))
            s[i] += v;
    }
    int query(int x) { // 查詢前綴和(有包含)
        int pre = 0;
        for (; x; x -= lowbit(x))
            pre += s[x];
        return pre;
    }
    Fenwick(vector<int> a) { // 建構資料結構
        n = a.size(); // a 是 0-based
        s.clear(), s.resize(n + 1, 0); // s 是 1-based
        for (int i = 1; i <= n; i++)
            update(i, a[i - 1]);
    }
};
// 使用:
Fenwick fw(a); // a is vector
Fenwick fw(n); // n is int
```

3.3 二維 BIT

```
struct BIT {
    int n;
    vector<vector<ll>> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) {
        bit.assign(n, vector<ll>(n, 0));
    }
    void update(int x, int y, ll val) { // 更新 (x, y)
        for (int i = x; i < n; i += lowbit(i))
```

```
        for (int j = y; j < n; j += lowbit(j))
            bit[i][j] += val;
    }
    ll query(int x, int y) { // 查詢 (1, 1) ~ (x, y) 的和
        ll ans = 0;
        for (int i = x; i; i -= lowbit(i))
            for (int j = y; j; j -= lowbit(j)) ans += bit[i][j];
        return ans;
    }
    ll range_query(int a, int b, int x, int y) {
        return query(x, y) - query(x, b - 1) -
            query(a - 1, y) + query(a - 1, b - 1);
    }
};
```

3.4 Seg/線段樹

```
struct Seg {
#define cl(x) (x << 1)
#define cr(x) (x << 1) + 1
    int _n;
    vector<int> f, tag;
    vector<int> arr;
    // Seg(int n) : _n(n), f(4 * n), tag(4 * n) {}
    Seg(vector<int> a) {
        _n = a.size();
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
        for (int i = 0; i < _n; i++) {
            arr[i] = a[i];
        }
        build(1, 0, _n - 1);
    }
    Seg(int x) {
        _n = x;
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
    }
    void build(int id, int l, int r) {
        if (l == r) {
            f[id] = arr[l];
            return;
        }
        int mid = (l + r) >> 1;
        build(cl(id), l, mid);
        build(cr(id), mid + 1, r);
        pull(id, l, r);
    }
    void pull(int id, int l, int r) {
        int mid = (l + r) >> 1;
        push(cl(id), l, mid);
        push(cr(id), mid + 1, r);
        f[id] = f[cl(id)] + f[cr(id)];
    }
    void push(int id, int l, int r) {
        if (tag[id]) {
            f[id] += tag[id] * (r - l + 1);
            if (l != r) {
                tag[cl(id)] += tag[id];
                tag[cr(id)] += tag[id];
            }
            tag[id] = 0;
        }
    }
    void update(int id, int l, int r, int x, int v) {
        if (l == r) {
            f[id] = v;
            return;
        }
        int mid = (l + r) >> 1;
        if (x <= mid) update(cl(id), l, mid, x, v);
        if (mid < x) update(cr(id), mid + 1, r, x, v);
```

```

    pull(id, l, r);
}

void update(int id, int l, int r, int ql, int qr,
            int v) {
    push(id, l, r);
    if (ql <= l and qr >= r) {
        tag[id] += v;
        return;
    }
    int mid = (l + r) >> 1;
    if (ql <= mid) {
        update(cl(id), l, mid, ql, qr, v);
    }
    if (qr > mid) {
        update(cr(id), mid + 1, r, ql, qr, v);
    }
    pull(id, l, r);
}

int query(int id, int l, int r, int sl, int sr) {
    push(id, l, r);
    if (sl <= l and sr >= r) {
        return f[id];
    }
    int res = 0;
    int mid = (l + r) >> 1;
    int ll = 0;
    int rr = 0;
    if (sl <= mid) {
        ll = query(cl(id), l, mid, sl, sr);
    }
    if (sr > mid) {
        rr = query(cr(id), mid + 1, r, sl, sr);
    }
    res = ll + rr;
    return res;
}
};

```

4 Graph

4.1 Topo/拓樸

```

vector<int> edge[MAXN], ans;
int deg[MAXN];

void topo(int n) {
    queue<int> q;

    for (int i = 1; i <= n; i++)
        if (!deg[i])
            q.push(i);

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        ans.push_back(u);

        for (int v : edge[u]) {
            deg[v]--;
            if (!deg[v])
                q.push(v);
        }
    }
}

```

4.2 bellman/邊長鬆弛

```

struct Edge {
    int from, to, weight;
} edge[MAXN];
int dis[MAXN];
void bellmen_ford(int n, int m) {
    dis[1] = 0; // 起點的距離一定是 0
    for (int j = 0; j < n - 1; j++) { // n 個節點要跑 n - 1 次
        for (int i = 0; i < m; i++) { // 對於所有邊都嘗試鬆弛
            if (dis[edge[i].to] >
                dis[edge[i].from] + edge[i].weight)

```

```

                dis[edge[i].to] =
                    dis[edge[i].from] + edge[i].weight;
            }
        }
    }
}

```

4.3 ola/歐拉

```

// 歐拉路徑(無向圖)兩個奇數點或者全偶點
// 歐拉路徑(有向圖)條件:起點out=in+1,終點in=out+1,其他點
// in=out
// 歐拉迴路(無向圖)所有點偶數點
// 歐拉迴路(有向圖)所有點in=out
vector<int> anse; // 邊
vector<int> ansp; // 點
vector<bool> vis(m, false);
function<void(int)> dfs=&](int u){
    while(!mp[u].empty()){
        auto [x,e]=mp[u].back();
        mp[u].pop_back();
        if(vis[e]){
            continue;
        }
        vis[e]=true;
        dfs(x);
        anse.pb(e);
    }
    ansp.pb(u);
};

```

4.4 LCA/最近公共祖先

```

struct{
    int n, logN;
    vector<vector<int>>> jump;
    vector<int> in, out;
    bool is_ancestor(int x, int y){
        if(x == -1 || y == -1) return 1;
        return in[x] <= in[y] && out[y] <= out[x];
    }
    int query(int x, int y){
        if(is_ancestor(x, y)) return x;
        if(is_ancestor(y, x)) return y;
        for(int i = logN - 1; i >= 0; i--){
            if(!is_ancestor(jump[x][i], y)) x = jump[x][i];
        }
        return jump[x][0];
    }
    void init(int _n, vector<vector<int>>>&v){ // size,
        adj array 0base
        n = _n;
        logN = log2(n) + 1;

        jump.resize(n, vector<int>(logN, -1));
        in.resize(n); out.resize(n);

        int timing = 1;
        auto dfs = [&](auto &&self, int x) -> void{
            in[x] = timing++;
            for(auto &i : v[x]){
                jump[i][0] = x;
                self(self, i);
            }
            out[x] = timing++;
            return;
        }; dfs(dfs, 0);

        for(int i = 1; i < logN; i++){
            for(int now = 1; now < n; now++){
                if(jump[now][i-1] == -1) continue;
                jump[now][i] = jump[jump[now][i-1]][i-1];
            }
        }
    }
}LCA;

```

4.5 km/二分圖最大權匹配

```

struct KM{ // max weight, for min negate the weights

```

```

int n, mx[MXN], my[MXN], pa[MXN];
ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
bool vx[MXN], vy[MXN];
void init(int _n) { // 1-based, N個節點
    n = _n;
    for(int i=1; i<=n; i++) fill(g[i], g[i]+n+1, 0);
}
void addEdge(int x, int y, ll w) {g[x][y] = w;} //左
    邊的集合節點x連邊右邊集合節點y權重為w
void augment(int y) {
    for(int x, z; y; y = z)
        x=pa[y], z=mx[x], my[y]=x, mx[x]=y;
}
void bfs(int st) {
    for(int i=1; i<=n; ++i) sy[i]=INF, vx[i]=vy[i]=0;
    queue<int> q; q.push(st);
    for(;;) {
        while(q.size()) {
            int x=q.front(); q.pop(); vx[x]=1;
            for(int y=1; y<=n; ++y) if(!vy[y]){
                ll t = lx[x]+ly[y]-g[x][y];
                if(t==0){
                    pa[y]=x;
                    if(!my[y]){augment(y);return;}
                    vy[y]=1, q.push(my[y]);
                }else if(sy[y]>t) pa[y]=x, sy[y]=t;
            }
        }
        ll cut = INF;
        for(int y=1; y<=n; ++y)
            if(!vy[y]&&cut>sy[y]) cut=sy[y];
        for(int j=1; j<=n; ++j){
            if(vx[j]) lx[j] -= cut;
            if(vy[j]) ly[j] += cut;
            else sy[j] -= cut;
        }
        for(int y=1; y<=n; ++y) if(!vy[y]&&sy[y]==0){
            if(!my[y]){augment(y);return;}
            vy[y]=1, q.push(my[y]);
        }
    }
}
ll solve(){ // 回傳值為完美匹配下的最大總權重
    fill(mx, mx+n+1, 0); fill(my, my+n+1, 0);
    fill(ly, ly+n+1, 0); fill(lx, lx+n+1, -INF);
    for(int x=1; x<=n; ++x) for(int y=1; y<=n; ++y) //
        1-base
        lx[x] = max(lx[x], g[x][y]);
    for(int x=1; x<=n; ++x) bfs(x);
    ll ans = 0;
    for(int y=1; y<=n; ++y) ans += g[my[y]][y];
    return ans;
} }graph;

```

4.6 Scc/強連通分量

```

//同一團u,v之間有路徑
//bln: 在哪个分輪 E原圖
struct Scc{
    int n, nScc, vst[MXN], bln[MXN];
    vector<int> E[MXN], rE[MXN], vec;
    void init(int _n){
        n = _n;
        for (int i=0; i<= n; i++)
            E[i].clear(), rE[i].clear();
    }
    void addEdge(int u, int v){
        E[u].PB(v); rE[v].PB(u);
    }
    void DFS(int u){
        vst[u]=1;
        for (auto v : E[u]) if (!vst[v]) DFS(v);
        vec.PB(u);
    }
    void rDFS(int u){
        vst[u] = 1; bln[u] = nScc;
        for (auto v : rE[u]) if (!vst[v]) rDFS(v);
    }
    void solve(){
        nScc = 0;
        vec.clear();
        fill(vst, vst+n+1, 0);
        for (int i=0; i<n; i++)
            if (!vst[i]) DFS(i);
    }
}

```

```

reverse(vec.begin(),vec.end());
fill(vst, vst+n+1, 0);
for (auto v : vec)
    if (!vst[v]){
        rDFS(v); nScc++;
    }
}
};

```

4.7 Bcc/雙連通分量

```

//雙連通分量定義:每個團塊刪除任意一個點都連通,每個割點
    都屬於多個雙連通分量
//割點出現數量>=2,割邊出現數量=1
struct BccVertex {
    int n, nScc, step, dfn[MXN], low[MXN];
    vector<int> E[MXN], sccv[MXN];
    int top, stk[MXN];
    void init(int _n) {
        n = _n;
        nScc = step = 0;
        for (int i = 0; i < n; i++) E[i].clear();
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        E[v].PB(u);
    }
    void DFS(int u, int f) {
        dfn[u] = low[u] = step++;
        stk[top++] = u;
        for (auto v : E[u]) {
            if (v == f) continue;
            if (dfn[v] == -1) {
                DFS(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] >= dfn[u]) {
                    int z;
                    sccv[nScc].clear();
                    do {
                        z = stk[--top];
                        sccv[nScc].PB(z);
                    } while (z != v);
                    sccv[nScc++].PB(u);
                }
            } else
                low[u] = min(low[u], dfn[v]);
        }
    }
    vector<vector<int>> solve() {
        vector<vector<int>> res;
        for (int i = 0; i < n; i++) dfn[i] = low[i] =
            -1;
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) {
                top = 0;
                DFS(i, i);
            }
        REP(i, nScc) res.PB(sccv[i]);
        return res;
    }
} graph;

```

5 dynamic programming

5.1 SOS DP

```

// 求子集和 或超集和 -> !(mask & (1 << i))
for(int i = 0; i<(1<<N); ++i) F[i] = A[i]; //預處理 狀
    態權重
for(int i = 0; i < N; ++i)
    for (int s = 0; s < (1<<N) ; ++s)
        if (s & (1 << i))
            F[s] += F[s ^ (1 << i)];

//窮舉子集合
for(int s = mask; s ; s = (s-1)&mask;)
//放這個做啥我不會用QAQ

```

6 Computational Geometry

6.1 default

```
int sgn(double x) { return (x > -eps) - (x < eps); }
template<class T>
struct pt{
    T x,y;
    pt(T _x,T _y):x(_x),y(_y){}
    pt():x(0),y(0){}
    pt operator * (T c){ return pt(x * c, y * c); }
    pt operator / (T c){ return pt(x / c, y / c); }
    pt operator + (pt a){ return pt(x + a.x, y + a.y); }
    pt operator - (pt a){ return pt(x - a.x, y - a.y); }
    T operator * (pt a){ return x * a.x + y * a.y; } //
        內積
    T operator ^ (pt a){ return x * a.y - y * a.x; } //
        外積
    bool operator < (pt a) const { return x < a.x || (x
        == a.x && y < a.y); } //按照x排
    bool operator==(const pt<T>& a) const { return x ==
        a.x && y == a.y; }
};
template<class T>
ld abs(pt<T> a) { //返回向量長度
    return sqrt((ld)(a.x * a.x + a.y * a.y));
}
template<class T>
struct Line { //直線
    pt<T> a, b;
    pt<T> dir()const{return b-a;}
};
template<class T> //線段
T ori(pt<T> a, pt<T> b, pt<T> c){ return (b - a) ^ (c -
    a); }
template<class T> //判斷點是否在線段上
bool PtOnSeg(pt<T> p, Line<T> L) {
    return sgn(ori(L.a, L.b, p)) == 0 and sgn((p - L.a)
        * (p - L.b)) <= 0;
}
```

6.2 極座標排序

```
//由+x軸往逆時針排序
struct point {
    int x, y;
};
long long cross(const point &a, const point &b) {
    return (long long) a.x * b.y - (long long) a.y * b.
        x;
}
bool cmp(const point &a, const point &b) {
    int ah = (a.y < 0 or (a.y == 0 and a.x < 0));
    int bh = (b.y < 0 or (b.y == 0 and b.x < 0));
    if (ah != bh) return ah < bh;
    return cross(a, b) > 0;
}
void argument_sort(vector<point> &points) {
    sort(points.begin(), points.end(), cmp);
}
```

6.3 ConvexHull/凸包

```
template<class T>
vector<pt<T>> Hull(vector<pt<T>> P) {
    sort(all(P));
    P.erase(unique(all(P)), P.end());
    P.insert(P.end(), P.rbegin() + 1, P.rend());
    vector<pt<T>> stk;
    for (auto p : P) {
        auto it = stk.rbegin();
        while (stk.rend() - it >= 2 and \
            ori(*next(it), *it, p) <= 0 and \
            (*next(it) < *it) == (*it < p)) {
            it++;
        }
        stk.resize(stk.rend() - it);
        stk.push_back(p);
    }
    stk.pop_back();
    return stk;
}
```

6.4 PtSegDist

```
template<class T> //計算點到線段的距離
ld PtSegDist(pt<T> p, Line<T> l) {
    ld ans = min(abs(p - l.a), abs(p - l.b));
    if (sgn(abs(l.a - l.b)) == 0) return ans;
    if (sgn((l.a - l.b) * (p - l.b)) < 0) return ans;
    // case1
    if (sgn((l.b - l.a) * (p - l.a)) < 0) return ans;
    // case1
    return min(ans, abs(ori(p, l.a, l.b)) / abs(l.a - l
        .b)); // case2
}
```

6.5 點是否在多邊形內

```
template<class T> //判斷點是否在多邊形內
int inPoly(pt<T> p, const vector<pt<T>> &P) {
    const int n = P.size();
    int cnt = 0;
    for (int i = 0; i < n; i++) {
        pt<T> a = P[i], b = P[(i + 1) % n];
        if (PtOnSeg(p, {a, b})) return 1; // on edge
        if ((sgn(a.y - p.y) == 1) ^ (sgn(b.y - p.y) ==
            1))
            cnt += sgn(ori(a, b, p));
    }
    return cnt == 0 ? 0 : 2; // out, in
}
```

6.6 最遠點對/選轉卡尺

```
double RotatingCalipers(vector<Pt> &v){ //逆時針
    int n=v.size();
    double dist=0;
    if(n==1) return 0;
    if(n==2) return dis(v[0]-v[1]);
    for(int i=0,j=2;i<n;i++){
        Pt a=v[i],b=v[(i+1)%n];
        while(cross(v[j],a,b)<cross(v[(j+1)%n],a,b))
            j=(j+1)%n;
        dist=max(dist,dis(v[j]-a));
        dist=max(dist,dis(v[j]-b));
    }
    return dist;
}
```

7 Flow

7.1 Dinic

```
struct Dinic{
    struct Edge{ int v,f,re; };
    int n,s,t,level[MXN];
    vector<Edge> E[MXN];
    void init(int _n, int _s, int _t){
        n = _n; s = _s; t = _t;
        for (int i=0; i<n; i++) E[i].clear();
    }
    void add_edge(int u, int v, int f){
        E[u].PB({v,f,SZ(E[v])});
        E[v].PB({u,0,SZ(E[u])-1});
    }
    bool BFS(){
        for (int i=0; i<n; i++) level[i] = -1;
        queue<int> que;
        que.push(s);
        level[s] = 0;
        while (!que.empty()){
            int u = que.front(); que.pop();
            for (auto it : E[u]){
                if (it.f > 0 && level[it.v] == -1){
                    level[it.v] = level[u]+1;
                    que.push(it.v);
                }
            }
            return level[t] != -1;
        }
    }
    int DFS(int u, int nf){
        if (u == t) return nf;
        int res = 0;
        for (auto &it : E[u]){
            if (it.f > 0 && level[it.v] == level[u]+1){

```

```

    int tf = DFS(it.v, min(nf,it.f));
    res += tf; nf -= tf; it.f -= tf;
    E[it.v][it.re].f += tf;
    if (nf == 0) return res;
} }
if (!res) level[u] = -1;
return res;
}
int flow(int res=0){
    while ( BFS() )
        res += DFS(s,2147483647);
    return res;
} }flow;
/*flow.init(n, source, sink); : 初始化點數、源點、匯點
flow.add_edge(u, v, w); : 加入一條點u連到點v容量為w的邊
flow.flow(); : 回傳值為最大流量
*/

```

8 String

8.1 KMP

```

int failure[MXN];
vector<int> KMP(vector<int>& t, vector<int>& p){
    vector<int> ret;
    if (p.size() > t.size()) return {};
    for (int i=1, j=failure[0]=-1; i<p.size(); ++i){
        while (j >= 0 && p[j+1] != p[i])
            j = failure[j];
        if (p[j+1] == p[i]) j++;
        failure[i] = j;
    }
    for (int i=0, j=-1; i<t.size(); ++i){
        while (j >= 0 && p[j+1] != t[i])
            j = failure[j];
        if (p[j+1] == t[i]) j++;
        if (j == p.size()-1){
            ret.push_back( i - p.size() + 1 );
            j = failure[j];
        }
    }
    return ret;
}
/*
vector<int> prefunc(const string& s){
    int n = s.size();
    vector<int> pi(n);
    for(int i=1,j=0;i<n;++i){
        j = pi[i-1];
        while(j && s[j] != s[i]) j = pi[j-1]; //往下找藍框
        if(s[j] == s[i]) ++j; //相等會多一格
        pi[i] = j;
    }
    return pi;
}
*/

```

8.2 Z-value

```

int z[MXN];
void Z_value(const string& s) {
    //z[i] = lcp(s[0...],s[i...])
    int i, j, left, right, len = s.size();
    left=right=0; z[0]=len;
    for(i=1;i<len;i++){
        j=max(min(z[i-left],right-i),0LL);
        for(;i+j<len&&s[i+j]==s[j];j++);
        z[i]=j;
        if(i+z[i]>right) {
            right=i+z[i];
            left=i;
        }
    }
}

```

8.3 馬拉車

```

//記得s.resize(2*n+1)
//z_value_pal(&s[0],n,a.data());
void z_value_pal(char *s,int len,int *z){
    len=(len<<1)+1;
    for(int i=len-1;i>=0;i--){
        s[i]=i&1?s[i>>1]:'@';
        z[0]=1;
        for(int i=1,l=0,r=0;i<len;i++){

```

```

            z[i]=i<r?min(z[l+l-i],r-i):1;
            while(i-z[i]>=0&&i+z[i]<len&&s[i-z[i]]==s[i+z[i]])
                ++z[i];
            if(i+z[i]>r) l=i,r=i+z[i];
        }
    }
}

```

8.4 minRotation/最小字典序旋轉

```

//rotate(begin(s),begin(s)+minRotation(s),end(s))
int minRotation(string s) {
    int a = 0, N = s.size(); s += s;
    rep(b,0,N) rep(k,0,N) {
        if(a+k == b || s[a+k] < s[b+k])
            {b += max(0LL, k-1LL); break;}
        if(s[a+k] > s[b+k]) {a = b; break;}
    } return a;
}

```

8.5 SAIS

```

// 此為後綴陣列模板
// 用法:
// 把要處理的字串轉成整數陣列傳入suffix_array(), 會得到
// SA與H維結果
const int N = 500010; // 有可能要在這就寫原本字串的兩
// 倍長
struct SA{
#define REP(i,n) for ( int i=0; i<int(n); i++ )
#define REP1(i,a,b) for ( int i=(a); i<=int(b); i++ )
    bool _t[N*2];
    int _s[N*2], _sa[N*2], _c[N*2], x[N], _p[N], _q[N*2],
        hei[N], r[N];
    int operator [] (int i){ return _sa[i]; }
    void build(int *s, int n, int m){
        memcpy(_s, s, sizeof(int) * n);
        sais(_s, _sa, _p, _q, _t, _c, n, m);
        mkhei(n);
    }
    void mkhei(int n){
        REP(i,n) r[_sa[i]] = i;
        hei[0] = 0;
        REP(i,n) if(r[i]) {
            int ans = i>0 ? max(hei[r[i-1]] - 1, 0) : 0;
            while(_s[i+ans] == _s[_sa[r[i]-1]+ans]) ans++;
            hei[r[i]] = ans;
        }
    }
    void sais(int *s, int *sa, int *p, int *q, bool *t,
        int *c, int n, int z){
        bool uniq = t[n-1] = true, neq;
        int nn = 0, nmzx = -1, *nsa = sa + n, *ns = s + n,
            lst = -1;
#define MS0(x,n) memset((x),0,n*sizeof(*(x)))
#define MAGIC(XD) MS0(sa, n); \
        memcpy(x, c, sizeof(int) * z); \
        XD; \
        memcpy(x + 1, c, sizeof(int) * (z - 1)); \
        REP(i,n) if(sa[i] && !t[sa[i]-1]) sa[x[sa[i]
            ]-1]++ = sa[i]-1; \
        memcpy(x, c, sizeof(int) * z); \
        for(int i = n - 1; i >= 0; i--) if(sa[i] && t[sa[i]
            ]-1) sa[--x[sa[i]-1]] = sa[i]-1;
        MS0(c, z);
        REP(i,n) uniq &= ++c[s[i]] < 2;
        REP(i,z-1) c[i+1] += c[i];
        if (uniq) { REP(i,n) sa[--c[s[i]]] = i; return; }
        for(int i = n - 2; i >= 0; i--) t[i] = (s[i]==s[i
            +1] ? t[i+1] : s[i]<s[i+1]);
        MAGIC(REP1(i,1,n-1) if(t[i] && !t[i-1]) sa[--x[s[i]
            ]]=p[q[i]=nn++]=i);
        REP(i, n) if (sa[i] && t[sa[i]] && !t[sa[i]-1]) {
            neq=lst<0||memcmp(s+sa[i],s+lst,(p[q[sa[i]]+1]-sa
                [i])*sizeof(int));
            ns[q[lst=sa[i]]]=nmzx+=neq;
        }
        sais(ns, nsa, p + nn, q + n, t + n, c + z, nn, nmzx
            + 1);
        MAGIC(for(int i = nn - 1; i >= 0; i--) sa[--x[s[p[
            nsa[i]]]] = p[nsa[i]]);
    }
}sa;

```

```

int H[ N ], SA[ N ]; // SA: 後綴陣列的length
// H[i]: SA[i]與SA[i -1]的共同子串
// 長(LCP)
void suffix_array(int* ip, int len) {
    // should padding a zero in the back
    // ip is int array, len is array length
    // ip[0..n-1] != 0, and ip[len] = 0
    ip[len++] = 0;
    sa.build(ip, len, 128);
    for (int i = 0; i < len; i++) {
        H[i] = sa.hei[i + 1];
        SA[i] = sa._sa[i + 1];
    }
    // resulting height, sa array \in [0,len)
}

```

8.6 Trie/字典樹

```

struct trie{
    trie *nxt[26];
    int cnt; //紀錄有多少個字串以此節點結尾
    int sz; //有多少字串的前綴包括此節點
    trie():cnt(0),sz(0){
        memset(nxt,0,sizeof(nxt));
    }
};

//創建新的字典樹
trie *root = new trie();
// 0(|s|)
void insert(string& s){
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        now->sz++;
        if(now->nxt[i-'a'] == NULL){
            now->nxt[i-'a'] = new trie();
        }
        now = now->nxt[i-'a']; //走到下一個字母
    }
    now->cnt++;
    now->sz++;
}

// 0(|s|)
int query_prefix(string& s){ //查詢有多少前綴為 s
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        if(now->nxt[i-'a'] == NULL){
            return 0;
        }
        now = now->nxt[i-'a'];
    }
    return now->sz;
}

int query_count(string& s){ //查詢字串 s 出現次數
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        if(now->nxt[i-'a'] == NULL){
            return 0;
        }
        now = now->nxt[i-'a'];
    }
    return now->cnt;
}

```

8.7 ACautomata/AC 自動機

```

//makefail before query
//用於多模式字串匹配
struct ACautomata{
    struct Node{
        int cnt,i;
        Node *go[26], *fail, *dic;
        Node (){
            cnt = 0; fail = 0; dic = 0; i = 0;
            memset(go,0,sizeof(go));
        }
    }pool[1048576],*root;
    int ans[1048576];
    int nMem,n_pattern;
    Node* new_Node(){
        pool[nMem] = Node();
    }
}

```

```

return &pool[nMem++];
}
void init() {
    nMem=0;root=new_Node();n_pattern=0;
    add("");
}
void add(const string &str) { insert(root,str,0); }
void insert(Node *cur, const string &str, int pos){
    for(int i=pos;i<str.size();i++){
        if(!cur->go[str[i]-'a']){
            cur->go[str[i]-'a'] = new_Node();
            cur=cur->go[str[i]-'a'];
        }
        cur->cnt++;
        cur->i=n_pattern++;
    }
}
void make_fail(){
    queue<Node*> que;
    que.push(root);
    while (!que.empty()){
        Node* fr=que.front(); que.pop();
        for (int i=0; i<26; i++){
            if (fr->go[i]){
                Node *ptr = fr->fail;
                while (ptr && !ptr->go[i]) ptr = ptr->fail;
                fr->go[i]->fail=ptr=(ptr?ptr->go[i]:root);
                fr->go[i]->dic=(ptr->cnt?ptr:ptr->dic);
                que.push(fr->go[i]);
            }
        }
    }
}
void query(string s){
    Node *cur=root;
    for(int i=0;i<(int)s.size();i++){
        while (cur&&!cur->go[s[i]-'a']) cur=cur->fail;
        cur=(cur?cur->go[s[i]-'a']:root);
        if(cur->i>=0) ans[cur->i]++;
        for(Node *tmp=cur->dic;tmp;tmp=tmp->dic)
            ans[tmp->i]++;
    } // ans[i] : number of occurrence of pattern i
}AC;

```

8.8 差分約束系統

- 2-SAT :
 - $(a_i \vee a_j) = \text{true} \quad \forall (i, j)$
 - 對於任意限制 $(x \vee y)$
 - 建兩條有向邊 (要多編號 $\neg x$)
 - $x \rightarrow \neg y$ and $y \rightarrow \neg x$
 - 跑 scc
 - $\text{scc.bl}[x] < \text{scc.bl}[\neg x] \Leftrightarrow x \text{ is true}$
 - $\text{scc.bl}[\neg x] < \text{scc.bl}[x] \Leftrightarrow x \text{ is false}$
 - $\exists x \text{ which } \text{scc.bl}[x] == \text{scc.bl}[\neg x] \Leftrightarrow \text{無解}$
- 差分約束:
 - n 個變數及 m 個約束條件
 - 求滿足所有 $x_j - x_i \leq b_k \quad (i, j \in [1, n], k \in [1, m])$
 - 的一組 $x_1 \dots x_n$
 - 可轉成 $x_j - x_i \leq b_k \rightarrow x_j \leq x_i + b_k$
 - 結論就是使得所有 x_j 變小以滿足上式
 - 建邊跑 SPFA/Bellman
 - 要多建起點 s 連到所有 i 且邊權 0, $\text{dis}[s] = 0$
 - 有負環則無解, 否則起點到所有 i 的距離為一組解
 - $x_j - x_i \leq k \Rightarrow \text{addEdge } i \xrightarrow{k} j$
 - $x_j - x_i \geq k \Rightarrow \text{addEdge } j \xrightarrow{-k} i$
 - $x_j = x_i \Rightarrow \text{addEdge } i \xrightarrow{0} j \text{ and } j \xrightarrow{0} i$

9 酷東西可能會用到

9.1 areaofrect/矩形聯集面積

```

struct AreaofRectangles{
#define cl(x) (x<<1)
#define cr(x) (x<<1|1)
    int n, id, sid;
    pair<int,int> tree[MXN<<3]; // count, area
    vector<int> ind;
    tuple<int,int,int,int> scan[MXN<<1];
    void point(int i, int l, int r){
        if(tree[i].first tree[i].second = ind[r+1] - ind[l]; //當前區間有被做加值
        else if(l != r){ //沒有就直接拿兩個子節點的第二個數字
            int mid = (l+r)>>1;

```

```

        tree[i].second = tree[cl(i)].second + tree[
            cr(i)].second;
    }
    else    tree[i].second = 0;
}
void upd(int i, int l, int r, int ql, int qr, int v
){
    if(ql <= l && r <= qr){
        tree[i].first += v;
        puint(i, l, r); return;
    }
    int mid = (l+r) >> 1;
    if(ql <= mid) upd(cl(i), l, mid, ql, qr, v);
    if(qr > mid) upd(cr(i), mid+1, r, ql, qr, v);
    puint(i, l, r);
}
void init(int _n){
    n = _n; id = sid = 0;
    ind.clear(); ind.resize(n<<1);
    fill(tree, tree+(n<<2), make_pair(0, 0));
}
void addRectangle(int lx, int ly, int rx, int ry){
    ind[id++] = lx; ind[id++] = rx;
    scan[sid++] = make_tuple(ly, 1, lx, rx);
    scan[sid++] = make_tuple(ry, -1, lx, rx);
}
int solve(){
    sort(ind.begin(), ind.end());
    ind.resize(unique(ind.begin(), ind.end()) - ind
        .begin()); //離散化
    sort(scan, scan + sid);
    int area = 0, pre = get<0>(scan[0]);
    for(int i = 0; i < sid; i++){
        auto [x, v, l, r] = scan[i];
        area += tree[l].second * (x-pre); //
        upd(l, 0, ind.size()-1, lower_bound(ind.
            begin(), ind.end(), l)-ind.begin(),
            lower_bound(ind.begin(), ind.end(), r)-
            ind.begin()-1, v);
        pre = x;
    }
    return area;
} }rect;

```

10 Math

10.1 basic

```

[a,b] 有包含
(a,b) 沒包含
ll mul(ll x,ll y,ll mod){
    ll ret=x*y-(ll)((long double)x/mod*y)*mod;
    // LL ret=x*y-(LL)((long double)x*y/mod+0.5)*mod;
    return ret<0?ret+mod:ret;
}
ll add(int x,int y,int mod){
    return (x+y)%mod;
}

```

10.2 快速幂

```

int ksm(int x, int y) {
    int ans = 1;
    while (y) {
        if (y & 1)
            ans = ans * x;
        x = x * x;
        y >>= 1;
    }
    return ans;
}

// 矩陣乘法
vector<vector<int>> mul(const vector<vector<int>> &A,
    const vector<vector<int>> &B) {
    if (A.empty() || B.empty() || A[0].empty() || B[0].
        empty()) { return {}; }
    int m = A.size();
    int k = A[0].size();
    int n = B[0].size();
    vector<vector<int>> C(m, vector<int>(n, 0));
    for (int i = 0; i < m; i++) {

```

```

        for (int p = 0; p < k; p++) if (A[i][p]) {
            for (int j = 0; j < n; j++) {
                C[i][j] = (C[i][j] + 1LL * A[i][p] * B[p][j])%MOD;
            }
        }
    }
    return C;
}

```

// 矩陣快速幂

```

vector<vector<int>> ksm(vector<vector<int>> base, int
    exp) {
    int n = base.size();
    vector<vector<int>> res(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) res[i][i] = 1;

    while (exp > 0) {
        if (exp & 1) res = mul(res, base);
        base = mul(base, base);
        exp >>= 1;
    }
    return res;
}

```

10.3 大數運算

//加法

```

string add(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        carry = 0;
    while (i >= 0 || j >= 0 || carry) {
        int sum = carry + (i >= 0 ? num1[i--] - '0' :
            0) + (j >= 0 ? num2[j--] - '0' : 0);
        carry = sum / 10;
        res += to_string(sum % 10);
    }
    reverse(res.begin(), res.end());
    return res;
}

```

//減法

```

string sub(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        borrow = 0;
    while (i >= 0) {
        int sub = (num1[i--] - '0') - (j >= 0 ? num2[j
            --] - '0' : 0) - borrow;
        if (sub < 0) {
            sub += 10;
            borrow = 1;
        } else {
            borrow = 0;
        }
        res += to_string(sub);
    }
    while (res.length() > 1 && res.back() == '0') res.
        pop_back(); // 去除前導0
    reverse(res.begin(), res.end());
    return res;
}

```

//乘法

```

string mul(string num1, string num2) {
    int n = num1.size(), m = num2.size();
    vector<int> res(n + m, 0);

    for (int i = n - 1; i >= 0; i--) {
        for (int j = m - 1; j >= 0; j--) {
            int p = (num1[i] - '0') * (num2[j] - '0');
            int sum = p + res[i + j + 1];

            res[i + j + 1] = sum % 10;
            res[i + j] += sum / 10;
        }
    }
    string ans = "";
    for (int x : res) {
        if (!ans.empty() && x == 0) // 跳過前導0
            ans += (char)(x + '0');
    }
}

```

```

    return ans.empty() ? "0" : ans;
}
//除法
bool smaller(string a, string b) {
    if (a.size() != b.size()) return a.size() < b.size();
    return a < b;
}
string div(string num1, string num2) {
    string cur = "", ans = "";
    for (char c : num1) {
        cur += c;
        // 去掉前導0
        while (cur.size() > 1 && cur[0] == '0') cur.erase(cur.begin());

        int x = 0;
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
            x++;
        }
        ans += (char)(x + '0');
    }
    // 去掉前導0
    while (ans.size() > 1 && ans[0] == '0') ans.erase(ans.begin());
    return ans;
}
string mod(string num1, string num2) { //餘數
    string cur = "";
    for (char c : num1) {
        cur += c;
        while (cur.size() > 1 && cur[0] == '0') cur.erase(cur.begin());
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
        }
    }
    return cur;
}

```

10.4 Prime

```

/* 12721, 13331, 14341, 75577, 123457, 222557, 556679
 * 999983, 1097774749, 1076767633, 100102021, 999997771
 * 1001010013, 1000512343, 987654361, 999991231
 * 999888733, 98789101, 987777733, 999991921, 1010101333
 * 1010102101, 1000000000039, 100000000000037
 * 2305843009213693951, 4611686018427387847
 * 9223372036854775783, 18446744073709551557 */
int mu[ N ], p_tbl[ N ];
vector<int> primes;
void sieve() {
    mu[ 1 ] = p_tbl[ 1 ] = 1;
    for( int i = 2 ; i < N ; i ++ ){
        if( !p_tbl[ i ] ){
            p_tbl[ i ] = i;
            primes.push_back( i );
            mu[ i ] = -1;
        }
        for( int p : primes ){
            int x = i * p;
            if( x >= M ) break;
            p_tbl[ x ] = p;
            mu[ x ] = -mu[ i ];
            if( i % p == 0 ){
                mu[ x ] = 0;
                break;
            }
        }
    }
}
vector<int> factor( int x ){
    vector<int> fac{ 1 };
    while( x > 1 ){
        int fn = SZ(fac), p = p_tbl[ x ], pos = 0;
        while( x % p == 0 ){
            x /= p;
            for( int i = 0 ; i < fn ; i ++ )
                fac.PB( fac[ pos ++ ] * p );
        }
    }
    return fac;
}

```

10.5 ExGCD

```

//ax+by=gcd(a,b)
PII gcd(int a, int b){
    if(b == 0) return {1, 0};
    PII q = gcd(b, a % b);
    return {q.second, q.first - q.second * (a / b)};
}

```

10.6 Phi

//歐拉函數 phi(n) 是指<=n中與n互質的數的個數

```

int _phi(int n) { // 0(sqrtN)
    int res = n, a = n;
    for (int i = 2; i * i <= a; i++) {
        if (a % i == 0) {
            res = res / i * (i - 1);
            while (a % i == 0) a /= i;
        }
    }
    if (a > 1) res = res / a * (a - 1);
    return res;
}

```

```

int phi[MXN]; // 建表 最大 1e7
void phi_table(int n) {
    phi[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (phi[i]) continue;
        for (int j = i; j <= n; j += i) {
            if (phi[j] == 0) phi[j] = j;
            phi[j] = phi[j] / i * (i - 1);
        }
    }
}

```

10.7 Euler/歐拉函數

```

int prime[350000]; ULL phi[5000005];
bool v[5000005];
#define N 5000000
void quick_euler() {
    int cnt = 0;
    for(int i = 2; i <= N; ++i) {
        if(!v[i]) prime[++cnt] = i, phi[i] = i - 1; //
        // 若 i 是質數，φ(i) = i - 1
        for(int j = 1; i * prime[j] <= N && j <= cnt; ++j) {
            v[i * prime[j]] = 1; // 不是質數就篩掉
            if(i % prime[j] == 0){
                phi[i * prime[j]] = phi[i] * prime[j];
                break;
            }
            else phi[i * prime[j]] = phi[i] * (prime[j] - 1); // 積性函數性質用在這裡
        }
    }
}

```

10.8 miller rabin/判斷質數

```

// n < 4,759,123,141      3 : 2, 7, 61
// n < 1,122,004,669,633  4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383  6 : pirmses <= 13
// n < 2^64                7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
// Make sure testing integer is in range [2, n-2] if
// you want to use magic.
ll magic[] = {};
bool witness(ll a, ll n, ll u, int t){
    if(!a) return 0;
    ll x = mypow(a, u, n); //ksm(x,y,mod)
    for(int i=0; i<t; i++){
        ll nx = mul(x, x, n);
        if(nx==1 && x!=1 && x!=n-1) return 1;
        x = nx;
    }
    return x!=1;
}
bool miller_rabin(ll n) {
    int s=(7);
    // iterate s times of witness on n
}

```

```

if(n<2) return 0;
if(!(n&1)) return n == 2;
ll u=n-1; int t=0;
// n-1 = u*2^t
while(!(u&1)) u>>=1, t++;
while(s--){
    ll a=magic[s]%n;
    if(witness(a,n,u,t)) return 0;
}
return 1;
}

```

10.9 Pollard Rho/質因數分解

```

//回傳n的其中一個因數
// does not work when n is prime 0(n^(1/4))
#define LL __int128
LL f(LL x, LL c, LL mod){ return add(mul(x,x,mod),c,mod); }
LL pollard_rho(LL n) {
    LL c = 1, x = 0, y = 0, p = 2, q, t = 0;
    while (t++ % 128 or gcd(p, n) == 1) {
        if (x == y) c++, y = f(x = 2, c, n);
        if (q = mul(p, abs(x-y), n)) p = q;
        x = f(x, c, n); y = f(f(y, c, n), c, n);
    }
    return gcd(p, n);
}
/*function<void(int)> fact=&](int x){
    if(x==1) return;
    if(miller_rabin(x)){
        f.push_back(x);return;}
    int tmp=pollard_rho(x);
    fact(tmp);fact(x/tmp);};
*/

```

10.10 FFT/傅立葉轉換 (多項式相乘)

```

const int MAXN = 262144<<1;
// (must be 2^k)
// before any usage, run pre_fft() first
typedef long double ld;
typedef complex<ld> cplx; //real() ,imag()
const ld PI = acos(-1);
const cplx I(0, 1);
cplx omega[MAXN+1];
void pre_fft(){
    for(int i=0; i<=MAXN; i++){
        omega[i] = exp(i * 2 * PI / MAXN * I);
    }
}
// n must be 2^k
void fft(int n, cplx a[], bool inv=false){
    int basic = MAXN / n;
    int theta = basic;
    for (int m = n; m >= 2; m >>= 1) {
        int mh = m >> 1;
        for (int i = 0; i < mh; i++) {
            cplx w = omega[inv ? MAXN-(i*theta%MAXN) : i*theta%MAXN];
            for (int j = i; j < n; j += m) {
                int k = j + mh;
                cplx x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
        }
        theta = (theta * 2) % MAXN;
    }
    int i = 0;
    for (int j = 1; j < n - 1; j++) {
        for (int k = n >> 1; k > (i ^ k); k >>= 1);
        if (j < i) swap(a[i], a[j]);
    }
    if(inv) for (i = 0; i < n; i++) a[i] /= n;
}
cplx arr[MAXN+1];
inline void mul(int _n, ll a[], int _m, ll b[], ll ans[]){
    int n=1, sum=_n+_m-1;
    while(n<sum)
        n<<=1;
    for(int i=0; i<n; i++){
        double x=(i<_n?a[i]:0), y=(i<_m?b[i]:0);
        arr[i]=complex<double>(x+y, x-y);
    }
}

```

```

}
fft(n, arr);
for(int i=0; i<n; i++){
    arr[i]=arr[i]*arr[i];
}
fft(n, arr, true);
for(int i=0; i<sum; i++){
    ans[i]=(ll)(arr[i].real()/4+0.5);
}

```

10.11 Theorem

- Lucas's Theorem :
For $n, m \in \mathbb{Z}^*$ and prime P , $C(m, n) \bmod P = \prod C(m_i, n_i)$ where m_i is the i -th digit of m in base P .
- Stirling approximation :
$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$
- Stirling Numbers(permutation $|P| = n$ with k cycles):
 $S(n, k) = \text{coefficient of } x^k \text{ in } \Pi_{i=0}^{n-1} (x+i)$
- Stirling Numbers(Partition n elements into k non-empty set):
$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$
- Pick's Theorem : $A = i + b/2 - 1$
 A : Area; i : grid number in the inner; b : grid number on the side
- Catalan number : $C_n = \binom{2n}{n} / (n+1)$
$$C_n^{n+m} - C_{n+1}^{n+m} = (m+n)! \frac{n-m+1}{n+1} \quad \text{for } n \geq m$$

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = 2 \binom{2n+1}{n+2} C_n$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = \sum_{i=0}^n C_i C_{n-i} \quad \text{for } n \geq 0$$
- Euler Characteristic:
planar graph: $V - E + F - C = 1$
convex polyhedron: $V - E + F = 2$
 V, E, F, C : number of vertices, edges, faces(regions), and components
- Kirchhoff's theorem :
 $A_{ii} = \deg(i), A_{ij} = (i, j) \in E ? -1 : 0$, Deleting any one row, one column, and cal the $\det(A)$
- Polya' theorem (c is number of color, m is the number of cycle size):
$$\left(\sum_{i=1}^m c^{\gcd(i, m)} \right) / m$$
- Burnside lemma:
$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$
- 錯排公式: (n 個人中, 每個人皆不再原來位置的組合數):
$$dp[0] = 1; dp[1] = 0;$$

$$dp[i] = (i-1) * (dp[i-1] + dp[i-2]);$$
- Bell 數 (有 n 個人, 把他們拆組的方法總數) :
$$B_0 = 1$$

$$B_n = \sum_{k=0}^n s(n, k) \quad (\text{second - stirling})$$

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$$
- Wilson's theorem :
$$(p-1)! \equiv -1 \pmod{p}$$
- Fermat's little theorem :
$$a^p \equiv a \pmod{p}$$
- Euler's totient function:
$$A^{B^C} \bmod p = \text{pow}(A, \text{pow}(B, C, p-1)) \bmod p$$
- 歐拉函數降幕公式:
$$A^B \bmod C = A^{B \bmod \phi(C) + \phi(C)} \bmod C$$
- 6 的倍數:
$$(a-1)^3 + (a+1)^3 + (-a)^3 + (-a)^3 = 6a$$
- Standard young tableau (標準楊表):
 $\lambda = (\lambda_1 \geq \dots \geq \lambda_k), \sum \lambda_i = n$ denoted by $\lambda \vdash n$
 $\lambda \vdash n$ 意思為 λ 整數拆分 n eg. $n = 10, \lambda = (6, 4)$ 此拆分可表示一種楊表形狀。
楊表: 第 1 列 λ_1 行 $\dots$ 第 k 列 λ_k 行的方格圖。
標準楊表: 每列從左到右遞增, 每行從上到下遞增。
Let T 為某一 Permutation 跑 RSK 後的標準楊表, 則此 Permutation 的 LDS、LIS 長度分別為 T 的列、行數。
- RSK Correspondence:
A permutation is bijective to (P, Q) 一對標準楊表
 P : Permutation 跑 RSK 算法的結果, 可為半標準楊表。
 Q : 可用來還原 Permutation (像排列矩陣)。

- Hook length formula (形狀為 λ 的標準楊表個數):

$$f^\lambda = \frac{n!}{\prod h_\lambda(i,j)}$$

$h_\lambda(i,j)$ = number of pair (x,y) where $(x=i \vee y=j) \wedge (x,y) \geq (i,j)$
且 (x,y) 落在形狀為 λ 的表上。

Recursion:

(i) $f^{(0,\dots,0)} = 1$

(ii) $f^{(\lambda_1,\dots,\lambda_m)} = \sum_{k=1}^m f^{(\lambda_1,\dots,\lambda_{k-1},\lambda_k-1,\lambda_{k+1},\dots,\lambda_m)}$

- 一個環上相鄰顏色不同的染色數量:

$$f(n,k) = (k-1)^n + (-1)^n (k-1)$$

n 是點的數量, k 是顏色的數量



為什麼要演奏春日影!

